

# Integration Manual

for S32K1\_S32M24X QDEC Driver

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## Chapter 1

### Revision History

| Revision | Date       | Author       | Description                                                  |
|----------|------------|--------------|--------------------------------------------------------------|
| 1.0      | 07.03.2025 | NXP RTD Team | S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0 |

## Chapter 2

### Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for QDEC Driver for S32K1\_S32M24X microcontrollers.

### 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116\_qfn32
- s32k116\_lqfp48
- s32k118\_lqfp48
- s32k118\_lqfp64
- s32k142\_lqfp48
- s32k142\_lqfp64
- s32k142\_lqfp100
- s32k142w\_lqfp48
- s32k142w\_lqfp64
- s32k144\_lqfp48
- s32k144\_lqfp64 / MWCT1014S\_lqfp64
- s32k144\_lqfp100 / MWCT1014S\_lqfp100
- s32k144\_mapbga100
- s32k144w\_lqfp48
- s32k144w\_lqfp64
- s32k146\_lqfp64

- s32k146\_lqfp100 / MWCT1015S\_lqfp100
- s32k146\_mapbga100 / MWCT1015S\_mapbga100
- s32k146\_lqfp144
- s32k148\_lqfp100
- s32k148\_mapbga100 / MWCT1016S\_mapbga100
- s32k148\_lqfp144
- s32k148\_lqfp176
- s32m241\_lqfp64
- s32m242\_lqfp64
- s32m243\_lqfp64
- s32m244\_lqfp64

All of the above microcontroller devices are collectively named as S32K1\_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

## 2.4 Acronyms and Definitions

| Term  | Definition                               |
|-------|------------------------------------------|
| API   | Application Programming Interface        |
| ASM   | Assembler                                |
| BSMI  | Basic Software Make file Interface       |
| C/CPP | C and C++ Source Code                    |
| CS    | Chip Select                              |
| CTU   | Cross Trigger Unit                       |
| DMA   | Direct Memory Access                     |
| DEM   | Diagnostic Event Manager                 |
| DET   | Development Error Tracer                 |
| ECU   | Electronic Control Unit                  |
| FIFO  | First In First Out                       |
| LSB   | Least Significant Bit                    |
| MCU   | Micro Controller Unit                    |
| MIDE  | Multi Integrated Development Environment |
| MSB   | Most Significant Bit                     |
| N/A   | Not Applicable                           |
| QDEC  | Quadrature Decoder                       |
| RAM   | Random Access Memory                     |
| SIU   | Systems Integration Unit                 |
| SWS   | Software Specification                   |
| VLE   | Variable Length Encoding                 |
| XML   | Extensible Markup Language               |

## 2.5 Reference List

| #  | Title                    | Version                                                 |
|----|--------------------------|---------------------------------------------------------|
| 1  | S32K1xx Reference Manual | Rev.14.1, 01/2024                                       |
| 2  | S32M24x Reference Manual | Rev. 5, 12/2024                                         |
| 3  | S32K1xx Data Sheet       | Rev. 14, 08/2021                                        |
| 4  | S32M2xx Data Sheet       | Rev. 7, 12/2024                                         |
| 5  | S32K116 Errata           | Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024 |
| 6  | S32K118 Errata           | Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024 |
| 7  | S32K142 Errata           | Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024 |
| 8  | S32K144 Errata           | Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024 |
| 9  | S32K144W Errata          | Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024 |
| 10 | S32K146 Errata           | Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024 |
| 11 | S32K148 Errata           | Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024 |
| 12 | S32M242 Errata           | Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024        |
| 13 | S32M244 Errata           | Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024        |

## Chapter 3

### Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

#### 3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [IAR Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 10.2.0 20200723 (Build 1728 Revision g5963bc8)
- IAR ANSI C/C++ Compiler V8.40.3.228/W32 for ARM Functional Safety
- Green Hills Multi 7.1.6d / Compiler 2020.1.4

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS\_T40D2M30I0R0 part of the plugin name is composed as follows:

- T = Target\_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative\_Id (e.g. D2 identifies S32K1 platform)
- M = SW\_Version\_Major and SW\_Version\_Minor
- I = SW\_Version\_Patch
- R = Reserved

##### 3.1.1 GCC Compiler/Assembler/Linker Options

###### 3.1.1.1 GCC Compiler Options

| Compiler Option | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| -mcpu=cortex-m4 | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices) |



| Compiler Option                       | Description                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -mcpu=cortex-m0plus                   | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                                                                                                 |
| -mthumb                               | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mlittle-endian                       | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -mfpv4-sp-d16                         | Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)                                                                                                                                     |
| -mfloat-abi=hard                      | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)                                                        |
| -mfpv4-sp-d16                         | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                                                                                                |
| -mfloat-abi=soft                      | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)                                                              |
| -std=c99                              | Specifies the ISO C99 base standard                                                                                                                                                                                                |
| -Os                                   | Optimize for size. Enables all -O2 optimizations except those that often increase code size                                                                                                                                        |
| -ggdb3                                | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -Wall                                 | Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros                                                 |
| -Wextra                               | This enables some extra warning flags that are not enabled by -Wall                                                                                                                                                                |
| -pedantic                             | Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option                                          |
| -Wstrict-prototypes                   | Warn if a function is declared or defined without specifying the argument types                                                                                                                                                    |
| -Wundef                               | Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero                                                                                                                          |
| -Wunused                              | Warn whenever a function, variable, label, value, macro is unused                                                                                                                                                                  |
| -Werror=implicit-function-declaration | Make the specified warning into an error. This option throws an error when a function is used before being declared                                                                                                                |
| -Wsign-compare                        | Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.                                                                                        |
| -Wdouble-promotion                    | Give a warning when a value of type float is implicitly promoted to double                                                                                                                                                         |
| -fno-short-enums                      | Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.                                                                                                                |
| -funsigned-char                       | Let the type char be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                              |
| -funsigned-bitfields                  | Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned                                                                                                                                |

| Compiler Option                 | Description                                                                                                                                                                                                                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -fomit-frame-pointer            | Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.                                                                                       |
| -fno-common                     | Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit |
| -fstack-usage                   | This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis                                                                                                             |
| -fdump-ipa-all                  | This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps                                                                                                                                                            |
| -c                              | Stop after assembly and produce an object file for each source file                                                                                                                                                                                                                   |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode                                                                                                                                         |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32↔M24x devices)                                                                                                  |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                                                                                                     |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPO↔RT as a macro, with definition 1. Allows drivers to be configured in user mode.                                                                                                                                                                 |
| -sysroot=                       | Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib                                                                                                                                                                                                          |
| -specs=nano.specs               | Use Newlib nano specs                                                                                                                                                                                                                                                                 |
| -specs=nosys.specs              | Do not use printf/scanf                                                                                                                                                                                                                                                               |

### 3.1.1.2 GCC Assembler Options

| Assembler Option     | Description                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Xassembler-with-cpp | Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)                           |
| -mcpu=cortex-m4      | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)                                                    |
| -mcpu=cortex-m0plus  | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                               |
| -mfpu=fpv4-sp-d16    | Specifies the floating-point hardware available on the target (for S32K14x devices)                                                                              |
| -mfloat-abi=hard     | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x devices) |
| -mfpu=auto           | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                              |

| Assembler Option | Description                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -mfloat-abi=soft | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices) |
| -mthumb          | Generates code that executes in Thumb state                                                                                                                           |
| -c               | Stop after assembly and produce an object file for each source file                                                                                                   |

### 3.1.1.3 GCC Linker Options

| Linker Option        | Description                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Wl,-Map,filename    | Produces a map file                                                                                                                                                                                                                |
| -T linkerfile        | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)                                                                                                                     |
| -entry=Reset_Handler | Specifies that the program entry point is Reset_Handler                                                                                                                                                                            |
| -nostartfiles        | Do not use the standard system startup files when linking                                                                                                                                                                          |
| -mcpu=cortex-m4      | Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)                                                                                                                      |
| -mcpu=cortex-m0plus  | Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)                                                                                                                                 |
| -mthumb              | Generates code that executes in Thumb state                                                                                                                                                                                        |
| -mfpu=fpv4-sp-d16    | Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)                                                                                                                                     |
| -mfloat-abi=hard     | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)                                                        |
| -mfpu=auto           | Specifies the floating-point hardware available on the target (for S32K11x devices)                                                                                                                                                |
| -mfloat-abi=soft     | Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)                                                              |
| -mlittle-endian      | Generate code for a processor running in little-endian mode                                                                                                                                                                        |
| -ggdb3               | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -lc                  | Link with the C library                                                                                                                                                                                                            |
| -lm                  | Link with the Math library                                                                                                                                                                                                         |
| -lgcc                | Link with the GCC library                                                                                                                                                                                                          |
| -n                   | Turn off page alignment of sections, and disable linking against shared libraries                                                                                                                                                  |
| -sysroot=            | Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib                                                                                                                                                       |
| -specs=nano.specs    | Use Newlib nano specs                                                                                                                                                                                                              |
| -specs=nosys.specs   | Do not use printf/scanf                                                                                                                                                                                                            |

### 3.1.2 IAR Compiler/Assembler/Linker Options

#### 3.1.2.1 IAR Compiler Options

| Compiler Option | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M4  | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices) |

| Compiler Option                 | Description                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M0+                 | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                                                                               |
| -cpu_mode=thumb                 | Generates code that executes in Thumb state                                                                                                                                                                      |
| -endian=little                  | Generate code for a processor running in little-endian mode                                                                                                                                                      |
| -fpu=FPv4-SP                    | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices)                                           |
| -fpu=none                       | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                                                                        |
| -e                              | Enables all IAR C language extensions                                                                                                                                                                            |
| -Ohz                            | Optimize for size. The compiler will emit AEABI attributes indicating the requested optimization goal. This information can be used by the linker to select smaller or faster variants of DLIB library functions |
| -debug                          | Makes the compiler include debugging information in the object modules. Including debug information will make the object files larger                                                                            |
| -no_clustering                  | Disables static clustering optimizations. Static and global variables defined within the same module will not be arranged so that variables that are accessed in the same function are close to each other       |
| -no_mem_idioms                  | Makes the compiler not optimize certain memory access patterns                                                                                                                                                   |
| -no_explicit_zero_opt           | Do not treat explicit initializations to zero of static variables as zero initializations                                                                                                                        |
| -require_prototypes             | Force the compiler to verify that all functions have proper prototypes. Generates an error otherwise                                                                                                             |
| -no_wrap_diagnostics            | Does not wrap long lines in diagnostic messages                                                                                                                                                                  |
| -diag_suppress=Pa050            | Suppresses diagnostic message Pa050                                                                                                                                                                              |
| -DIAR                           | Predefine IAR as a macro, with definition 1                                                                                                                                                                      |
| -DUSE_SW_VECTOR_MODE            | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.                                                                   |
| -DI_CACHE_ENABLE                | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                              |
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                                |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.                                                                                             |

### 3.1.2.2 IAR Assembler Options

| Assembler Option | Description                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=Cortex-M4   | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)                                               |
| -cpu=Cortex-M0+  | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                          |
| -fpu=FPv4-SP     | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x devices) |
| -fpu=none        | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                   |
| -cpu_mode thumb  | Selects the thumb mode for the assembler directive CODE                                                                                                     |
| -g               | Disables the automatic search for system include files                                                                                                      |
| -r               | Generates debug information                                                                                                                                 |

### 3.1.2.3 IAR Linker Options

| Linker Option                | Description                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -map filename                | Produces a map file                                                                                                                                                    |
| -config linkerfile           | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)                                                         |
| -cpu=Cortex-M4               | Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)                                                          |
| -cpu=Cortex-M0+              | Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)                                                                     |
| -fpu=FPv4-SP                 | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices) |
| -fpu=none                    | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)                              |
| -entry __start               | Treats __start as a root symbol and start label                                                                                                                        |
| -enable_stack_usage          | Enables stack usage analysis. If a linker map file is produced, a stack usage chapter is included in the map file                                                      |
| -skip_dynamic_initialization | Dynamic initialization (typically initialization of C++ objects with static storage duration) will not be performed automatically during application startup           |
| -no_wrap_diagnostics         | Does not wrap long lines in diagnostic messages                                                                                                                        |

## 3.1.3 GHS Compiler/Assembler/Linker Options

### 3.1.3.1 GHS Compiler Options

| Compiler Option   | Description                                                              |
|-------------------|--------------------------------------------------------------------------|
| -cpu=cortexm4     | Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices) |
| -cpu=cortexm0plus | Selects target processor: Arm Cortex M0+ (for S32K11x devices)           |
| -thumb            | Selects generating code that executes in Thumb state                     |

| Compiler Option            | Description                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -fpu=vfpv4_d16             | Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x or S32M24x devices)                                                          |
| -fsingle                   | Use hardware single-precision, software double-precision FP instructions (for S32K14x or S32M24x devices)                                                                                                                      |
| -fsoft                     | Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices) |
| -C99                       | Use (strict ISO) C99 standard (without extensions)                                                                                                                                                                             |
| -ghstd=last                | Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)                                                                    |
| -Osize                     | Optimize for size                                                                                                                                                                                                              |
| -gnu_asm                   | Enables GNU extended asm syntax support                                                                                                                                                                                        |
| -dual_debug                | Generate DWARF 2.0 debug information                                                                                                                                                                                           |
| -G                         | Generate debug information                                                                                                                                                                                                     |
| -keeptempfiles             | Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory                   |
| -Wimplicit-int             | Produce warnings if functions are assumed to return int                                                                                                                                                                        |
| -Wshadow                   | Produce warnings if variables are shadowed                                                                                                                                                                                     |
| -Wtrigraphs                | Produce warnings if trigraphs are detected                                                                                                                                                                                     |
| -Wundef                    | Produce a warning if undefined identifiers are used in #if preprocessor statements                                                                                                                                             |
| -unsigned_chars            | Let the type char be unsigned, like unsigned char                                                                                                                                                                              |
| -unsigned_fields           | Bitfields declared with an integer type are unsigned                                                                                                                                                                           |
| -no_commons                | Allocates uninitialized global variables to a section and initializes them to zero at program startup                                                                                                                          |
| -no_exceptions             | Disables C++ support for exception handling                                                                                                                                                                                    |
| -no_slash_comment          | C++ style // comments are not accepted and generate errors                                                                                                                                                                     |
| -prototype_errors          | Controls the treatment of functions referenced or called when no prototype has been provided                                                                                                                                   |
| -incorrect_pragma_warnings | Controls the treatment of valid #pragma directives that use the wrong syntax                                                                                                                                                   |
| -c                         | Stop after assembly and produce an object file for each source file                                                                                                                                                            |
| -DUSE_SW_VECTOR_MODE       | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode                                                                                  |
| -DI_CACHE_ENABLE           | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)                                            |

| Compiler Option                 | Description                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -DENABLE_FPU                    | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices) |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode                                               |

### 3.1.3.2 GHS Assembler Options

| Assembler Option           | Description                                                                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -cpu=cortexm4              | Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices)                                                                                                                                                       |
| -cpu=cortexm0plus          | Selects target processor: Arm Cortex M0+ (for S32K11x devices)                                                                                                                                                                 |
| -fpu=vfpv4_d16             | Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x devices)                                                                     |
| -fsingle                   | Use hardware single-precision, software double-precision FP instructions (for S32K14x devices)                                                                                                                                 |
| -fsoft                     | Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices) |
| -preprocess_assembly_files | Controls whether assembly files with standard extensions such as .s and .asm are preprocessed                                                                                                                                  |
| -list                      | Creates a listing by using the name and directory of the object file with the .lst extension                                                                                                                                   |
| -c                         | Stop after assembly and produce an object file for each source file                                                                                                                                                            |

### 3.1.3.3 GHS Linker Options

| Linker Option            | Description                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -e Reset_Handler         | Make the symbol Reset_Handler be treated as a root symbol and the start label of the application                                                                                                  |
| -T linker_script_file.ld | Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)                                                                         |
| -map                     | Produce a map file                                                                                                                                                                                |
| -keepmap                 | Controls the retention of the map file in the event of a link error                                                                                                                               |
| -Mn                      | Generates a listing of symbols sorted alphabetically/numerically by address                                                                                                                       |
| -delete                  | Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them |
| -ignore_debug_references | Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers                  |
| -Llibrary_path           | Points to library_path (the libraries location) for thumb2 to be used for linking                                                                                                                 |
| -larch                   | Link architecture specific library                                                                                                                                                                |
| -lstartup                | Link run-time environment startup routines. The source code for the modules in this library is provided in the src/libstartup directory                                                           |

| Linker Option           | Description                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -lind_sd                | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K14x or S32M24x devices) |
| -lind_sf                | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K11x devices)            |
| -v                      | Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols                                                                                                                     |
| -keep=C40_Ip_AccessCode | Avoid linker remove function C40_Ip_AccessCode from Fls module because it is not referenced explicitly                                                                                                                                              |
| -nostartfiles           | Controls the start files to be linked into the executable                                                                                                                                                                                           |

### 3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the QDEC driver for S32K1 microcontrollers. To avoid integration of incompatible files, all the include files from other modules shall have the same AR\_MAJOR\_VERSION and AR\_MINOR\_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

#### QDEC files:

- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_Types.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_EnvCfg.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_Ipw.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_Ipw\_Irq.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_Ipw\_Types.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Ftm\_Qdec\_Ip.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Ftm\_Qdec\_Ip\_Types.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Ftm\_Qdec\_Irq\_Ip.h
- ..\Qdec\_TS\_T40D2M30I0R0\src\Qdec.c
- ..\Qdec\_TS\_T40D2M30I0R0\src\Qdec\_Ipw.c
- ..\Qdec\_TS\_T40D2M30I0R0\src\Ftm\_Qdec\_Ip.c
- ..\Qdec\_TS\_T40D2M30I0R0\src\Ftm\_Qdec\_Irq\_Ip.c

#### QDEC generated files (these files should be generated by the user using a configuration tool):

- Qdec\_Cfg.h
- Ftm\_Qdec\_Ip\_Cfg.h
- Ftm\_Qdec\_Ip\_CfgDefines.h
- Qdec\_Ipw\_PBcfg.h
- Qdec\_PBcfg.h
- Ftm\_Qdec\_Ip\_PBcfg.h
- Qdec\_PBcfg.c



- Qdec\_Ipw\_PBcfg.c
- Ftm\_Qdec\_Ip\_PBcfg.c

#### Files from Base common folder:

- ..\Base\_TS\_T40D2M30I0R0\include\BasicTypes.h
- ..\Base\_TS\_T40D2M30I0R0\include\Compiler.h
- ..\Base\_TS\_T40D2M30I0R0\include\CompilerDefinition.h
- ..\Base\_TS\_T40D2M30I0R0\include\Compiler\_Cfg.h
- ..\Base\_TS\_T40D2M30I0R0\include\Devassert.h
- ..\Base\_TS\_T40D2M30I0R0\include\Qdec\_MemMap.h
- ..\Base\_TS\_T40D2M30I0R0\include\StandardTypes.h
- ..\Base\_TS\_T40D2M30I0R0\generate\_PC\include\Mcal.h
- ..\Base\_TS\_T40D2M30I0R0\header\[DerivativeName].h
- ..\Base\_TS\_T40D2M30I0R0\header\[DerivativeName]\_FTM.h

For S32K1 DerivativeName is one in list: S32K116, S32K118, S32K142, S32K142W, S32K144, S32K144W, S32K146, S32K148, S32M241 S32M242 S32M243 S32M244

#### Files from Det folder:

- ..\Det\_TS\_T40D2M30I0R0\include\Det.h

## 3.3 Setting up the plugins

The QDEC driver was designed to be configured by using the EB Tresos Studio (version EB tresos Studio 29.0.0 or later).

### 3.3.1 Location of various files inside the Qdec module folder

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
  - ..\Qdec\_TS\_T40D2M30I0R0\config\Qdec.xdm
  - ..\EcuC\_TS\_T40D2M30I0R0\config\EcuC.xdm
  - ..\Os\_TS\_T40D2M30I0R0\config\Os.xdm
  - ..\Base\_TS\_T40D2M30I0R0\config\Base.xdm
  - ..\Resource\_TS\_T40D2M30I0R0\config\Resource.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
  - ..\Qdec\_TS\_T40D2M30I0R0\autosar\Qdec.epd
  - ..\EcuC\_TS\_T40D2M30I0R0\autosar\EcuC.epd
  - ..\Os\_TS\_T40D2M30I0R0\autosar\Os.epd
  - ..\Base\_TS\_T40D2M30I0R0\autosar\Base.epd
  - ..\Resource\_TS\_T40D2M30I0R0\autosar\Resource.epd
- Code Generation Templates for Pre-Compile time configuration parameters:
  - ..\Qdec\_TS\_T40D2M30I0R0\include\Qdec\_Cfg.h

- ..\Qdec\_TS\_T40D2M30I0R0\include\Ftm\_Qdec\_Ip\_Cfg.h
- ..\Qdec\_TS\_T40D2M30I0R0\include\Ftm\_Qdec\_Ip\_CfgDefines.h

### 3.3.2 Dependencies

- RESOURCE is required to select processor derivative. Current Qdec driver has support for the following derivatives, each one having attached a Resource file: s32k116\_qfn32, s32k116\_lqfp48, s32k118\_lqfp48, s32k118\_lqfp64, s32k142\_lqfp48, s32k142\_lqfp64, s32k142\_lqfp100, s32k142w\_lqfp48, s32k142w\_lqfp64, s32k144\_lqfp48, s32k144\_lqfp64, s32k144\_lqfp100, s32k144\_mapbga100, s32k144w\_lqfp48, s32k144w\_lqfp64, s32k146\_lqfp64, s32k146\_lqfp100, s32k146\_mapbga100, s32k146\_lqfp144, s32k148\_lqfp100, s32k148\_mapbga100, s32k148\_lqfp144, s32k148\_lqfp176, s32m241\_lqfp64, s32m242\_lqfp64, s32m243\_lqfp64, s32m244\_lqfp64
- DET is required for signalling the development error detection (parameters out of range, null pointers, etc).
- BASE is required for Qdec specific header files and other header definitions.



## Chapter 4

### Function calls to module

- [Function Calls during Start-up](#)
- [Function Calls during Shutdown](#)
- [Function Calls during Wake-up](#)

#### 4.1 Function Calls during Start-up

None.

#### 4.2 Function Calls during Shutdown

None.

#### 4.3 Function Calls during Wake-up

None.

## Chapter 5

### Module requirements

- Exclusive areas to be defined in BSW scheduler
- Exclusive areas not available on this platform
- Peripheral Hardware Requirements
- ISR to configure within AutosarOS - dependencies
- ISR Macro
- Other AUTOSAR modules - dependencies
- Data Cache Restrictions
- User Mode support
- MultiPartition Support

#### 5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, Qdec is using the services of Schedule Manager (SchM) for entering and exiting the exclusive areas. The following critical regions are used in the Qdec driver:

**Exclusive Areas are used in High level driver layer (HLD)**

**QDEC\_EXCLUSIVE\_AREA\_01** is used in function Qdec\_GetState to protect the updates for:

- FTM\_SC register
- FTM\_QDCTRL register
- FTM\_CNT register

**QDEC\_EXCLUSIVE\_AREA\_02** is used in function Qdec\_SetClockMode to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_03** is used in function Qdec\_CounterState to protect the updates for:

- FTM\_MODE register
- FTM\_SC register
- FTM\_QDCTRL register

**QDEC\_EXCLUSIVE\_AREA\_04** is used in function Qdec\_ResetState to protect the updates for:

- FTM\_SC register

- FTM\_CNT register

**QDEC\_EXCLUSIVE\_AREA\_05** is used in function Qdec\_DisableOverFlowNotification to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_06** is used in function Qdec\_EnableOverFlowNotification to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_07** is used in function Qdec\_SetMode to protect the updates for:

- FTM\_MODE register
- FTM\_SC register
- FTM\_QDCTRL register

**QDEC\_EXCLUSIVE\_AREA\_08** is used in function ISR(FTM\_1\_ISR) to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_08** is used in function ISR(FTM\_2\_ISR) to protect the updates for:

- FTM\_SC register

**Exclusive Areas are implemented in Low level driver layer (IPL)**

**QDEC\_EXCLUSIVE\_AREA\_01** is used in function Ftm\_Qdec\_Ip\_GetState to protect the updates for:

- FTM\_SC register
- FTM\_QDCTRL register
- FTM\_CNT register

**QDEC\_EXCLUSIVE\_AREA\_02** is used in function Ftm\_Qdec\_Ip\_SetClockMode to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_03** is used in function Ftm\_Qdec\_Ip\_CounterState to protect the updates for:

- FTM\_MODE register
- FTM\_SC register
- FTM\_QDCTRL register

**QDEC\_EXCLUSIVE\_AREA\_04** is used in function Ftm\_Qdec\_Ip\_ResetState to protect the updates for:

- FTM\_SC register
- FTM\_CNT register

**QDEC\_EXCLUSIVE\_AREA\_05** is used in function Ftm\_Qdec\_Ip\_DisableOverFlowNotification to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_06** is used in function Ftm\_Qdec\_Ip\_EnableOverFlowNotification to protect the updates for:

- FTM\_SC register

**QDEC\_EXCLUSIVE\_AREA\_07** is used in function Ftm\_Qdec\_Ip\_SetMode to protect the updates for:

## Module requirements

- FTM\_MODE register
- FTM\_SC register
- FTM\_QDCTRL register

| Exclusive Area Matrix  |     |     |                        |                        |                        |                        |                        |                        |                        |     |
|------------------------|-----|-----|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-----|
| Exclusive Area ID      | N/A | N/A | QDEC_EXCLUSIVE_AREA_01 | QDEC_EXCLUSIVE_AREA_02 | QDEC_EXCLUSIVE_AREA_03 | QDEC_EXCLUSIVE_AREA_04 | QDEC_EXCLUSIVE_AREA_05 | QDEC_EXCLUSIVE_AREA_06 | QDEC_EXCLUSIVE_AREA_07 | N/A |
| QDEC_EXCLUSIVE_AREA_01 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_02 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_03 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_04 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_05 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_06 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |
| QDEC_EXCLUSIVE_AREA_07 |     |     | x                      | x                      | x                      | x                      | x                      | x                      | x                      |     |

Figure 5.1 Exclusive Areas

The critical regions from interrupts are grouped in "Interrupt Service Routines Critical Regions (composed diagram)". If an exclusive area is "exclusive" with the composed "Interrupt Service Routines Critical Regions (composed diagram)" group, it means that it is exclusive with each one of the ISR critical regions.

## 5.2 Exclusive areas not available on this platform

List of exclusive areas which are not available on this platform (or blank if they're all available).

- None

## 5.3 Peripheral Hardware Requirements

None.

## 5.4 ISR to configure within AutosarOS - dependencies

The following ISR's are used by the QDEC driver: The ISR table is presented below. Depending on the derivative used, some of the ISRs may not be available. For complete details please consult the Reference Manual:

For M24x:

| ISR Name       | HW INT Vector | Observations                               |
|----------------|---------------|--------------------------------------------|
| ISR(FTM_1_ISR) | 110           | FTM1 Counter overflow and Reload interrupt |
| ISR(FTM_2_ISR) | 116           | FTM2 Counter overflow and Reload interrupt |

For S32K11x:

| ISR Name       | HW INT Vector | Observations                               |
|----------------|---------------|--------------------------------------------|
| ISR(FTM_1_ISR) | 17            | FTM1 Counter overflow and Reload interrupt |

For S32K14x:

| ISR Name       | HW INT Vector | Observations                               |
|----------------|---------------|--------------------------------------------|
| ISR(FTM_1_ISR) | 110           | FTM1 Counter overflow and Reload interrupt |
| ISR(FTM_2_ISR) | 116           | FTM2 Counter overflow and Reload interrupt |

## 5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

### 5.5.1 Without an Operating System

The macro *USING\_OS\_AUTOSAROS* must not be defined.

#### 5.5.1.1 Using Software Vector Mode

The macro *USE\_SW\_VECTOR\_MODE* must be defined and the ISR macro is defined as: *#define ISR(IsrName) void IsrName(void)*

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

#### 5.5.1.2 Using Hardware Vector Mode

The macro *USE\_SW\_VECTOR\_MODE* must not be defined and the ISR macro is defined as: *#define ISR(IsrName) INTERRUPT\_FUNC void IsrName(void)*

In this case, the drivers' interrupt handlers must also handle the context save and restore.

### 5.5.2 With an Operating System

Please refer to your OS documentation for description of the ISR macro.

## 5.6 Other AUTOSAR modules - dependencies

- **BASE:** The BASE module contains the common files/definitions needed by all MCAL modules.
- **DET - Development Error Tracer:** This module is necessary for enabling Development error detection. The API function used is *Det\_ReportError()*. Activation of Development error detection is configurable using 'QdecDevErrorDetect' configuration parameter.
- **DEM - Diagnostic Event Manager:** This module is necessary for enabling reporting of production relevant error status. It is not used with current QDEC implementation as the production relevant error codes are not present.
- **EcuC module:** This module is necessary for handling PostBuild Variant. It allows users to configure multiple configuration.
- **RESOURCE:** The RESOURCE module is used to select microcontroller's derivatives.

## 5.7 Data Cache Restrictions

None.

## 5.8 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

### 5.8.1 User Mode configuration in the module

The QDEC driver can be run in user mode.

### 5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may have the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_Ip_TrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED (parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to be visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_Ip_TrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

## 5.9 MultiPartition Support

1. The Qdec implements the "MCAL Distribution" according to type II, in which the mappable element is set to instance for FTM Ip. For additional details, please refer to `AUTOSAR_EXP_BSWDistributionGuide`.
2. The Qdec and the mappable elements can be allocated to zero, one or several ECUC partitions, by means of "`QdecEcucPartionRef`". If the Qdec is mapped to zero ECUC partitions, the Qdec behavior reverts to single-core implementation, similar to previous Autosar versions. If the Qdec is mapped to one or more ECUC partitions, the Qdec enforces the following multi-core assumptions:



- The Qdec assumes there are 2 cases, a single EcucPartition allocated per core and multipartitions allocated to 1 core. Internally, the module will use the partition Id returned by GetUserId API to reference the appropriate global data and configuration elements.
- The Qdec assumes the EcucCoreIDs are defined in a compact/consecutive order, starting from zero. The rationale is that the number of EcucPartitions is used for dimensioning the Qdec internal variables and the EcucCoreIDs are used for indexing those variables.
- The Qdec assumes that initialization is performed on each partition, Qdec\_Init() is called separately for each partition, using a different configuration structure. (Type II)
- The Qdec initialization expects the upper layer will pass the correct initialization pointer, specific to the partition in which the driver is to be used. For example: EcucPartition\_1 is assigned to partition 1; Qdec\_Init function will be called with Qdec\_Config\_EcucPartition\_1 configuration structure, on partition 1.
- The Qdec will check upon each API call if the requested resource is configured to be available on the current partition, if DET error reporting is enabled.
- The Qdec requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multipartition context.
- The Qdec assumes that RTE module implements the EXCLUSIVE AREAS to be partition-aware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different partitions already, thus implementing the EXCLUSIVE AREAS in a blocking manner (ex: spin-lock) on a multipartition scope, might affect the performance of the drivers on the two partitions, although they might access separate HW elements.
- The Qdec assumes that each interrupt is routed by the system only to the partition on which is supposed to be serviced.
- The configuration structure name shall be available in the caller scope of Qdec\_Init function by being declared with EXTERN, according to its generated name.

#### Module specific limitation:

- For current implementation, Qdec driver does not support instance mapping with zero ECUC partition in configuration. Therefore the driver will force user to map each instance with one partition only.



## Chapter 6

### Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API Requirements](#)
- [Calls to Notification Functions, Callbacks, Callouts](#)

#### 6.1 Main function calls within BSW scheduler

None.

#### 6.2 API Requirements

None.

#### 6.3 Calls to Notification Functions, Callbacks, Callouts

None.

## Chapter 7

### Memory allocation

- Sections to be defined in Qdec\_MemMap.h
- Linker command file

#### 7.1 Sections to be defined in Qdec\_MemMap.h

| Section name                                 | Type of section    | Description                                                                                                                                                                                                          |
|----------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QDEC_START_SEC_CONFIG_DATA↔<br>A_UNSPECIFIED | Configuration Data | Start of Memory Section for Config Data                                                                                                                                                                              |
| QDEC_STOP_SEC_CONFIG_DATA↔<br>_UNSPECIFIED   | Configuration Data | End of memory Section for Config Data                                                                                                                                                                                |
| QDEC_START_SEC_CODE                          | Code               | Start of memory Section for Code                                                                                                                                                                                     |
| QDEC_STOP_SEC_CODE                           | Code               | End of memory Section for Code                                                                                                                                                                                       |
| QDEC_START_SEC_VAR_INIT_UN↔<br>SPECIFIED     | Variables          | Start of memory Section for Variables                                                                                                                                                                                |
| QDEC_STOP_SEC_VAR_INIT_UN↔<br>PECIFIED       | Variables          | End of memory Section for Variables                                                                                                                                                                                  |
| QDEC_START_SEC_VAR_CLEARE↔<br>D_UNSPECIFIED  | Variables          | Start of memory Section for Variables. Used for variables, constants, structure, array and unions when SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. For instance used for variables of unknown size |
| QDEC_STOP_SEC_VAR_CLEARED↔<br>_UNSPECIFIED   | Variables          | End of memory Section for Variables. Used for variables, constants, structure, array and unions when SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. For instance used for variables of unknown size   |

#### 7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>"\_MemMap.h.



## Chapter 8

### Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

## Chapter 9

### External assumptions for driver

The section presents requirements that must be complied with when integrating the QDEC driver into the application.

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EA_RTD_00071               | If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used.                                                                                                                                                                                                                                                                     |
| EA_RTD_00081               | The integrator shall assure that <MSN>_Init() and <MSN>_DeInit() functions do not interrupt each other.                                                                                                                                                                                                                                                                |
| EA_RTD_00082               | When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: <b>Rationale:</b> This ensures that no other buffers/variables compete for the same cache lines.                                                                            |
| EA_RTD_00092               | The integrator shall allocate a single EcucPartition per core or the partition in which the Qdec is allocated shall be exclusively mapped to a core. Note: Internally, the Qdec will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements, that is why a core should reference only one configured partition. |
| EA_RTD_00093               | The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero.                                                                                                                                                                                                                                                                           |
| EA_RTD_00094               | When multicore support is enabled, the application shall call Qdec_Init() for each core, using the dedicated configuration pointer for that core.                                                                                                                                                                                                                      |
| EA_RTD_00096               | The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used.                                                                                                                                                                                                                                            |
| EA_RTD_00106               | Standalone IP configuration and HL configuration of the same driver shall be done in the same project                                                                                                                                                                                                                                                                  |
| EA_RTD_00107               | The integrator shall use the IP interface only for hardware resources that were configured for standalone IP usage. Note: The integrator shall not directly use the IP interface for hardware resources that were allocated to be used in HL context.                                                                                                                  |
| EA_RTD_00108               | The integrator shall use the IP interface to build a CDD, therefore the BSWMD will not contain reference to the IP interface                                                                                                                                                                                                                                           |

## External assumptions for driver

| External Assumption Req ID | External Assumption Text                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EA_RTD_00113               | When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention Call<Function_Name>TRUSTED(parameter1,parameter2,...) in Integration/User code. They need to be visible in Os.h for the driver to call them. They will call RTD <Function_Name>() as trusted functions in OS specific manner(). |

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